EXP: 14 Write a code using RAW
DATE: 9/10/26 Sockets to implement packet Sniffing

AIM:

To capture and analyse network packets and display their protocols, source IP address, and destination IP addresses.

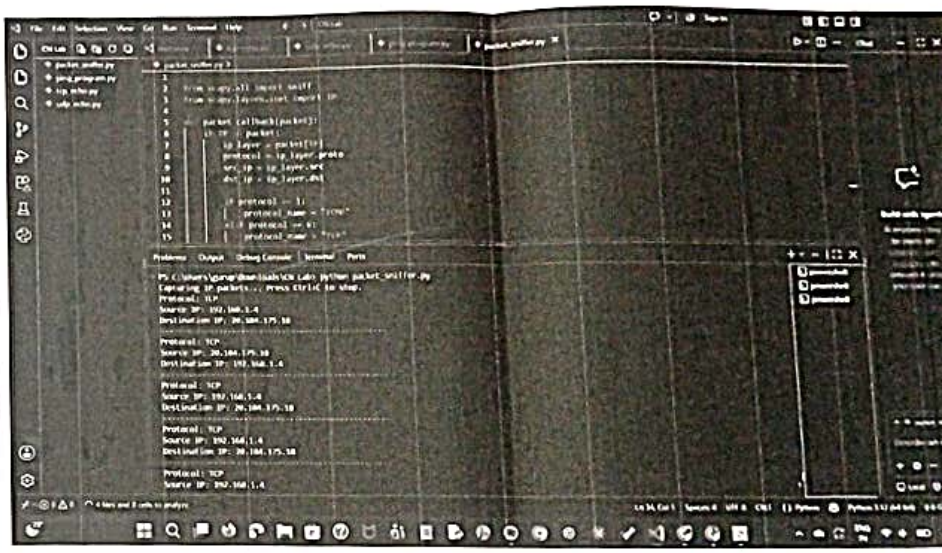
PROGRAM:

Packet_sniffer.py

```
from scalpy.all import sniff
from scalpy.layers.inet import IP
def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
        else:
            protocol_name = "Unknown
Protocol"
```

```
        print("Protocol:", protocol_name)
        print("Source IP:", src_ip)
        print("Destination IP:", dst_ip)
        print("-" * 50)

if __name__ == "__main__":
    print("Capturing IP packets...")
Press ctrl+c to stop.")
    sniff(
        iface = None,
        prn = packet_callback,
        filter = "ip",
        store = False
    )
```



RESULT:

The packet-sniffing program was successfully implemented.